University of Florence, Italy
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To: The Editors-in-Chief, *Empirical Software Engineering*

Manuscript: *Deep Green AI: Energy Efficiency of Deep Learning across Language–Framework Ecosystems*

Dear Editors-in-Chief,

We submit the above manuscript for consideration as a regular article in *Empirical Software Engineering*.

What the paper contributes. It is a controlled, replicated comparison of the energy cost of deep-learning training and inference across seven language–framework ecosystems: Python with PyTorch, TensorFlow and JAX; C++ with LibTorch; Java with Deeplearning4j; R with `torch`; and Rust with `tch`. The campaign is 210 runs on one GPU host. A written specification defines what "the same experiment" means across stacks, and 110 executable checks enforce it against the source and the built binaries. Every measurement block is read twice, by hardware energy counters and by a widely used software estimator.

Training energy differs across ecosystems by 7.4×–9.8×, and by 1.1×–1.6× among the three stacks that share one exported model and backend build. A declared contrast cell of 70 runs at 224×224 shows the spread narrowing within the LibTorch lineage (1.3×–2.8×) and widening across all seven stacks (15.3×–19.1×). The estimator agrees with the counters on energy to 0.3 %, but its reported duration understates derived power by up to 13.0×. The paper also includes a catalogue of the defect classes we found, our own among them.

Why EMSE. The paper is an empirical software-engineering study before it is a machine-learning one. Its central methodological claim is that a cross-ecosystem comparison is only as good as the specification and enforcement behind it. It treats the measurement apparatus as an object of study, reports between-run uncertainty from independent replications, and states its threats to validity in the terms the field uses. It also contributes to Green software engineering. The study is built for open science: every number in the manuscript is generated from the raw records by a public pipeline, except the few the supplement lists (Online Resource 1, Sections C.5 and D.3), and one command rebuilds the paper.

Prior submission. An earlier version of this study was submitted to the *Journal of Systems and Software* and rejected. We took the reviews as a reason to redo the study rather than to revise the text:

- a new measurement campaign on new hardware;
- a written experiment specification with an automated conformance checker;
- dual-instrument measurement of every block;
- the accelerator-saturation contrast cell;
- an audit of the defects in the earlier work.

The earlier reviews and our point-by-point account of how each was addressed are in the replication repository (`REVIEWERS_RESPONSE.md`), and we will supply them directly on request.

Data and code availability. The implementations, the specification, the conformance checker, the measurement bridge, the raw per-block records of the campaign and of the contrast cell, and the analysis pipeline are public at `https://github.com/Pampaj7/DeepGreen`. We will deposit an archived copy with a DOI before publication.

Declarations. The manuscript is not under consideration elsewhere and has not been published. All authors have read and approved the submission. We declare no competing interests.

Yours sincerely,

Leonardo Pampaloni and Marco Pagliocca (corresponding authors), Enrico Vicario, Roberto Verdecchia
University of Florence, Italy